University of Michigan
Using Python to Access Web Data

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Using Python to Access Web Data

by University of Michigan

About this Course

This course will show how one can treat the Internet as a source of data. We will scrape, parse, and read web data as well as access data using web APIs. We will work with HTML, XML, and JSON data formats in Python. This course will cover Chapters 11-13 of the textbook "Python for Everybody". To succeed in this course, you should be familiar with the material covered in Chapters 1-10 of the textbook and the first two courses in this specialization. These topics include variables and expressions, conditional execution (loops, branching, and try/except), functions, Python data structures (strings, lists, dictionaries, and tuples), and manipulating files. This course covers Python 3.

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Taught by:

Charles Russell Severance
Clinical Professor, School of Information

Basic Info	Course 3 of 5 in the Python for Everybody Specialization
Level	Beginner
Commitment	6 weeks of study, 2-4 hours/week
Language	English Subtitles: Arabic, French, Bengali, Uzbek, Ukrainian, Chinese (Simplified), Greek, Italian, Portuguese (Brazil), Vietnamese, Dutch, Korean, Oriya, German, Pashto, Urdu, Russian, Thai, Indonesian, Swedish, Turkish, Azerbaijani, Spanish, Dari, Hindi, Japanese, Kazakh, Hungarian, Polish
How To Pass	Pass all graded assignments to complete the course.
User Ratings	★ 4.8

Syllabus

Module 1

Getting Started

In this section you will install Python and a text editor. In previous classes in the specialization this was an optional assignment, but in this class it is the first requirement to get started. From this point forward we will stop using the browser-based Python grading environment because the browser-based Python environment (Skulpt) is not capable of running the more complex programs we will be developing in this class.

6 videos, 5 readings

1. Video: Welcome to The Course
2. Reading: Python Textbook
3. Reading: Help us learn more about you!
4. Reading: Course Syllabus
5. Video: Welcome to Python - Guido van Rossum
6. Reading: Notes on Choice of Text Editor
7. Reading: Notice for Auditing Learners: Assignment Submission
8. Video: Windows 10: Installing Python and Writing A Program
9. Video: Windows: Taking Screen Shots
10. Video: Macintosh: Using Python and Writing A Program
11. Video: Macintosh: Taking Screen Shots
12. App Item: Check External Tool Grades

Graded: Peer Review: Installing and Running Python Screen Shots

View Less

Module 2

Chapter Eleven: Regular Expressions

Regular expressions are a very specialized language that allow us to succinctly search strings and extract data from strings. Regular expressions are a language unto themselves. It is not essential to know how to use regular expressions, but they can be quite useful and powerful.

4 videos, 1 reading

1. Video: Regular Expressions
2. Video: Extracting Data
3. Reading: Python Regular Expression Quick Guide
4. Video: Bonus: Office Hours - Den Haag
5. Video: Bonus Interview: Bjarne Stroustrup - C++

Graded: Regular Expressions
Graded: Extracting Data With Regular Expressions

Module 3

Chapter Twelve: Networks and Sockets

In this section we learn about the protocols that web browsers use to retrieve documents and web applications use to interact with Application Program Interfaces (APIs).

8 videos, 1 reading

1. Video: Networked Technology
2. Video: Hypertext Transfer Protocol (HTTP)
3. Video: Worked Example: Sockets
4. Video: Using the Developer Console to Explore HTTP
5. Reading: If You Want to Learn More
6. Video: Bonus: Leonard Kleinrock - The First Two Packets on the ARPANET
7. Video: Bonus Video: Robert Cailliau - co-Inventor of the Web
8. Video: Bonus: Office Hours - Atlanta GA (Buckhead)
9. Video: Fun: Dr. Chuck @ CNN Reading the News

Graded: Networks and Sockets
Graded: Understanding the Request / Response Cycle

Module 4

Chapter Twelve: Programs that Surf the Web

In this section we learn to use Python to retrieve data from web sites and APIs over the Internet.

8 videos, 1 reading

1. Video: Unicode Characters and Strings
2. Video: Retrieving Web Pages
3. Video: Worked Example: Using Urllib
4. Video: Parsing Web Pages
5. Video: Worked Example: BeautifulSoup
6. Reading: Notes Regarding the Use of BeautifulSoup
7. Video: Bonus: Office Hours - Montreal
8. Video: Bonus Interview: Tim Berners-Lee - Inventing the Web
9. Video: Fun: I Got My Mojo Working - Geneva, Switzerland

Graded: Reading Web Data From Python
Graded: Scraping HTML Data with BeautifulSoup
Graded: Following Links in HTML Using BeautifulSoup

Module 5

Chapter Thirteen: Web Services and XML

In this section, we learn how to retrieve and parse XML (eXtensible Markup Language) data.

8 videos

1. Video: Data on the Web
2. Video: eXtensible Markup Language (XML)
3. Video: XML Schema
4. Video: Parsing XML
5. Video: Worked Example: XML
6. Video: Interview: Roy Fielding - Understanding the REST Architecture
7. Video: Bonus: Office Hours - Boston
8. Video: Bonus Video: Ian Horrocks / RDF / OWL (Advanced)

Graded: eXtensible Markup Language
Graded: Extracting Data from XML

Module 6

Chapter Thirteen: JSON and the REST Architecture

In this module, we work with Application Program Interfaces / Web Services using the JavaScript Object Notation (JSON) data format.

10 videos, 2 readings

1. Video: JavaScript Object Notation (JSON)
2. Video: Worked Example: JSON
3. Video: Interview: Douglas Crockford - Discovering JSON
4. Video: Service Oriented Approach
5. Video: Video: Service Oriented Architectures
6. Video: Using Application Programming Interfaces
7. Video: Worked Example: OpenStreetMap API
8. Video: Bonus: Office Hours - Melbourne, AU
9. Video: Bonus: Office Hours - Santa Monica, CA
10. Video: Bonus: Class Reunion at Bletchley Park
11. Reading: Please Rate this Course on Class-Central
12. Reading: Post-Course Survey

Graded: REST, JSON, and APIs
Graded: Extracting Data from JSON
Graded: Using a Geo Location API

How It Works

General

What do start dates and end dates mean?

Once you enroll, you'll have access to all videos, readings, quizzes, and programming assignments (if applicable). If you choose to explore the content without purchasing, you may not be able to access certain assignments. If you don't finish all graded assignments before the end of the session, you can reset your deadlines. Your progress will be saved and you'll be able to pick up where you left off.

What are due dates? Is there a penalty for submitting my work after a due date?

Within a course, there are suggested due dates to help you manage your schedule and keep work from piling up. Quizzes and programming assignments can be submitted late without consequence. However, it is possible that you won't receive a grade if you submit your peer-graded assignment too late because classmates usually review assignment within three days of the assignment deadline.

Can I re-attempt an assignment?

Yes. If you want to improve your grade, you can always try again. If you're re-attempting a peer-graded assignment, re-submit your work as soon as you can to make sure there's enough time for your classmates to review your work. In some cases you may need to wait before re-submitting a programming assignment or quiz. We encourage you to review learning material during this delay.

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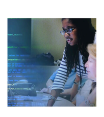
Course 3 of Specialization

Learn to Program and Analyze Data with Python

Develop programs to gather, clean, analyze, and visualize data.



University of Michigan

Python for Everybody

View the course in catalog

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